Chapter 11

11. Multifactor designs with no interactions.

1. Two-factor ANOVA.
2. Embedding into LM framework.
3. Testing.
4. Sums of square decomposition.
5. Data examlpe.
6. Randomized blocks.
7. The Latin square.

Two-factor ANOVA

We have used single-factor ANOVA to test for equality of r different population means or the mean associated to r different levels of some factor.

Often there is more than one factor of interest and we will now be considering a multifactor ANOVA. We start with two factors (factor A and factor B, say).

We will assume for the rest of this course a balanced design, i.e. the number of observations related to each factor combination is the same. In this particular chapter we consider the situation when there is only 1 observation per factor combination.

Two-factor ANOVA

Example

We wish to test $r = 3$ fabrics from erasable pens with $s = 4$ washing treatments.

	y	pen	wash
1	0.97	1	1
2	0.48	1	2
3	0.48	1	3
4	0.46	1	4
5	0.77	2	1
6	0.14	2	2
7	0.22	2	3
8	0.25	2	4
9	0.67	3	1
10	0.39	3	2
11	0.57	3	3
12	0.19	3	4

Two-factor ANOVA

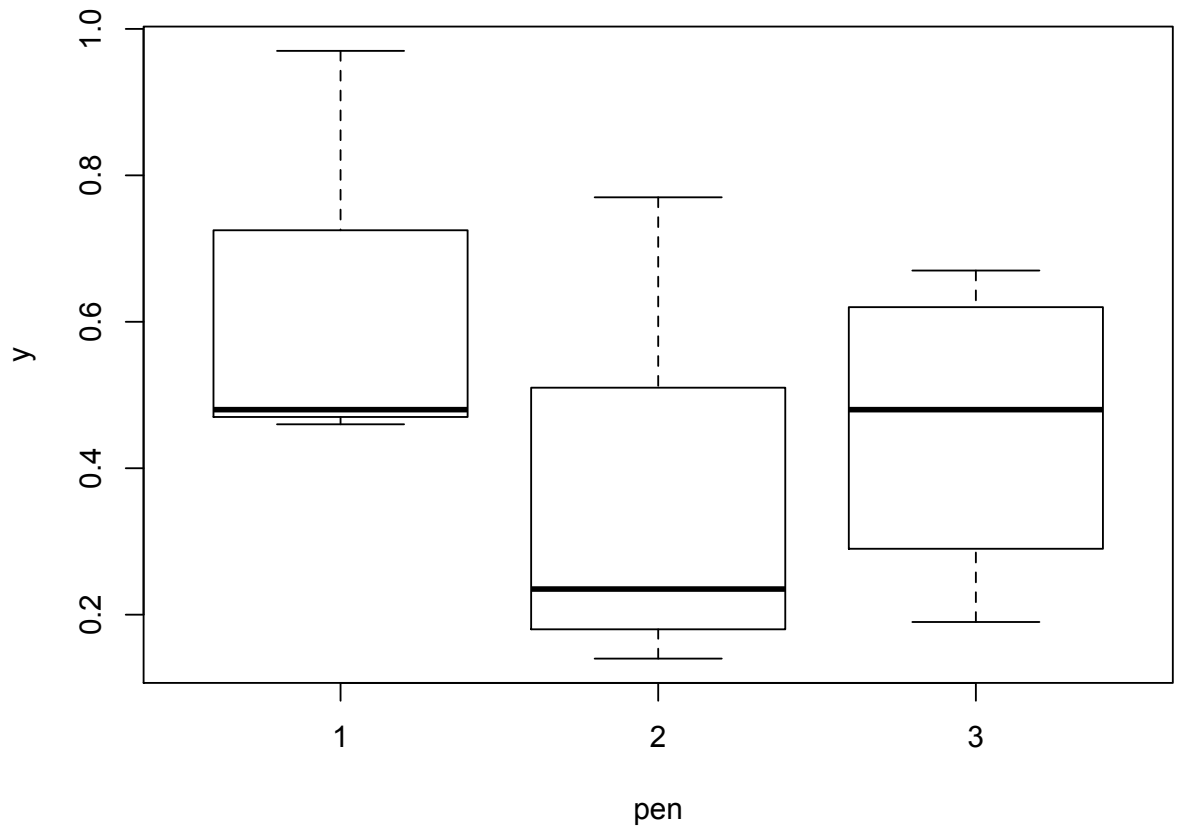
The response Y_{ij} $1 \leq i \leq r$ and $1 \leq j \leq s$ are quantitative indicators. The smaller, the better.

		Wash treatment					
		1	2	3	4	Total	Average
Pen	1	.97	.48	.48	.46	2.39	.598
	2	.77	.14	.22	.25	1.38	.345
	3	.67	.39	.57	.19	1.82	.455
	Total	2.41	1.01	1.27	.90	5.59	
	Average	.803	.337	.423	.300		.466

Question: does the type of pen or the type of wash treatment have an impact on the results?

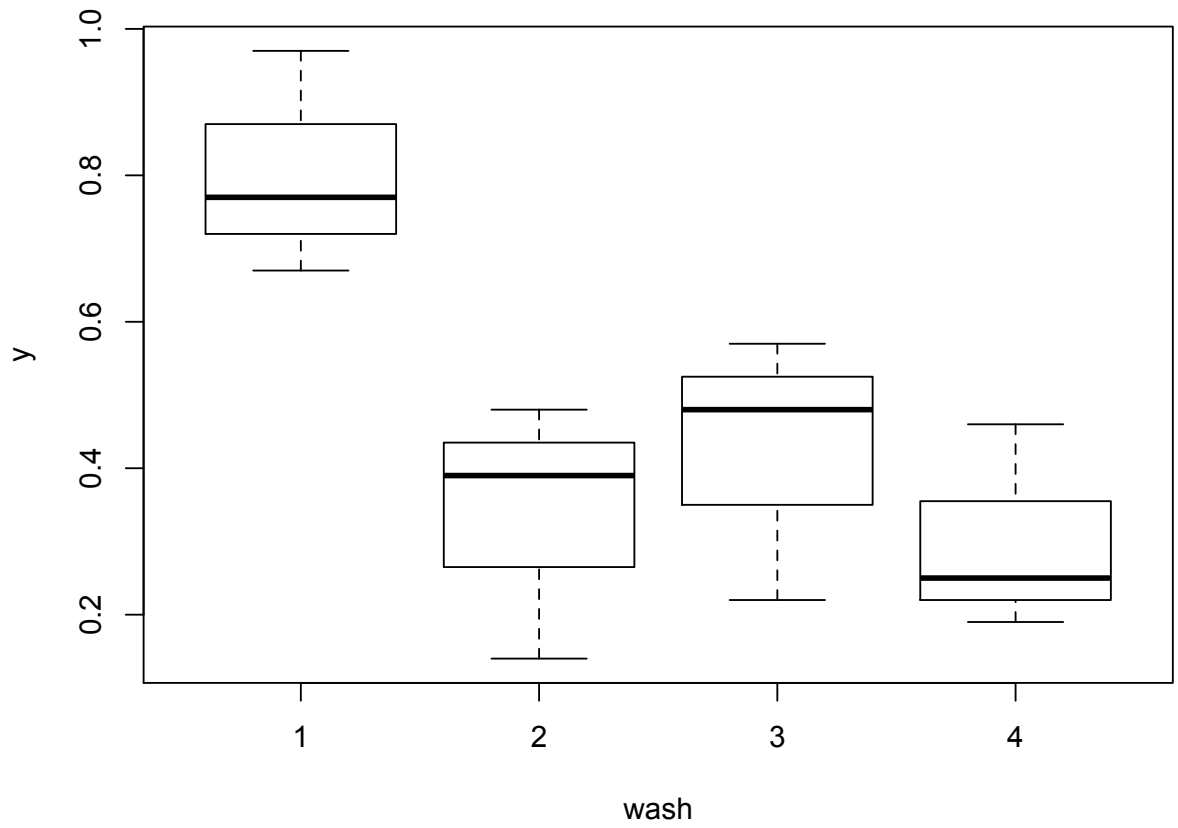
Two-factor ANOVA

```
> boxplot(y~pen)
```



Two-factor ANOVA

```
> boxplot(y~wash)
```



Two-factor ANOVA

We begin by stating the model:

$$Y_{ij} = \mu_{ij} + \epsilon_{ij}, \quad 1 \leq i \leq r, \quad 1 \leq j \leq s.$$

Since there is only one observation per factor combination, we don't need a triple index. We have sample size $n = rs$.

The boxplots indicate that homogeneous variance assumption is reasonable.

As usual we are going to assume that the errors are iid normal:

$$\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

Without any further constraints on μ_{ij} we have $n + 1$ parameters $\Longrightarrow$ the model is over-parametrized.

Two-factor ANOVA

To overcome the problem of over-parametrisation we impose an additive model

$$H_{A+B}: \mu_{ij} = \alpha_i + \beta_j.$$

Thus under H_{A+B} the data vector

$$Y = (Y_{11}, \ldots, Y_{1s}, \ldots, Y_{r1}, \ldots, Y_{rs})'$$

has mean

$$\mu = (\alpha_1 + \beta_1, \ldots, \alpha_1 + \beta_s, \ldots, \alpha_r + \beta_1, \ldots, \alpha_r + \beta_s)'.$$

The linear space L_{A+B} related to H_{A+B} has dimension $r + s - 1$ (will be shown below). So the model is depending on $r + s$ parameters, whereas we have $n = rs$ observations.

Two-factor ANOVA

Due to the additivity assumption, this model allows for no interactions!

For example, if $\alpha_2 > \alpha_1$ it means that for every j the mean μ_{2j} must be larger than μ_{1j}.

We should first check if this assumption is realistic at all. To this end we consider a so-called interaction plot. These are line charts

$$\ell_i : \text{line}\big((j, \mu_{ij}) \colon 1 \leq j \leq s\big) \quad \text{for} \quad i = 1, \ldots, r.$$

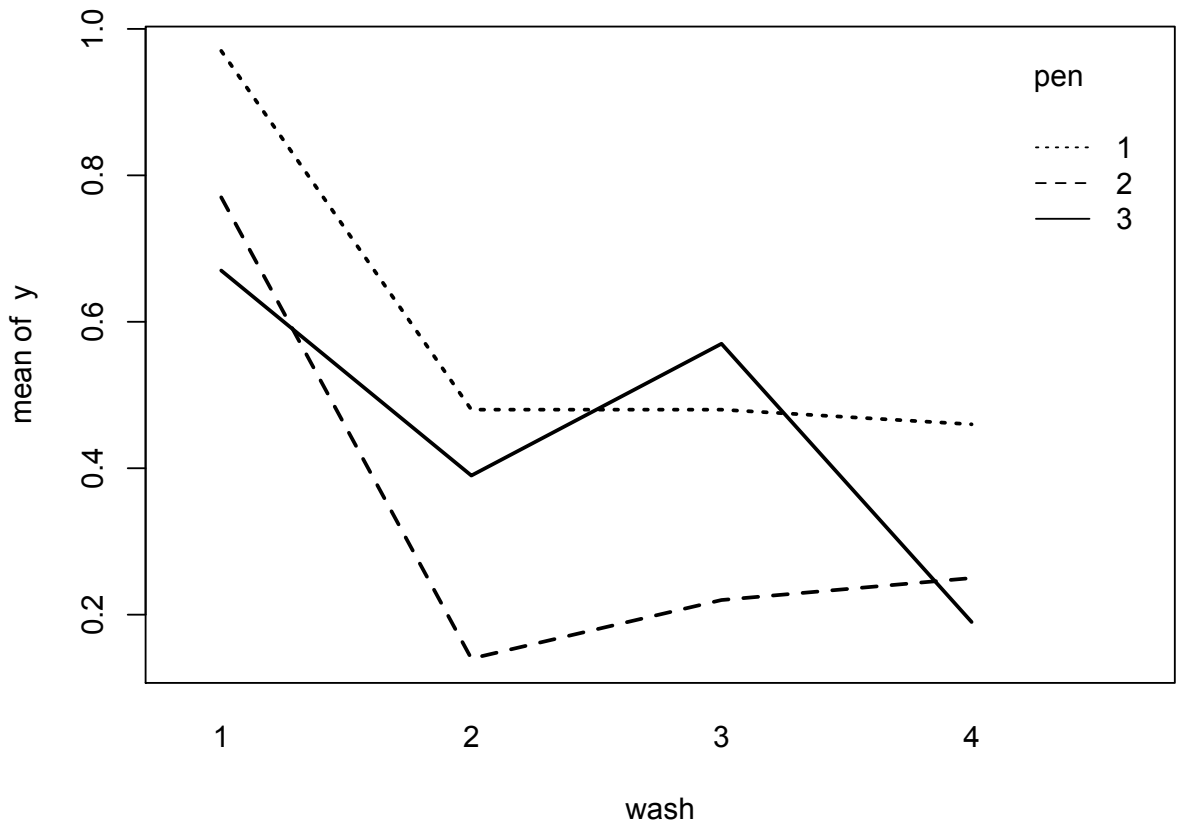
(Of course, we can change the role of i and j.)

Since $\mu_{ij} - \mu_{i'j} = \alpha_i - \alpha_{i'}$, the lines ℓ_i need to be parallel.

In applications, we have to replace μ_{ij} by Y_{ij} and require that these lines are approximately parallel.

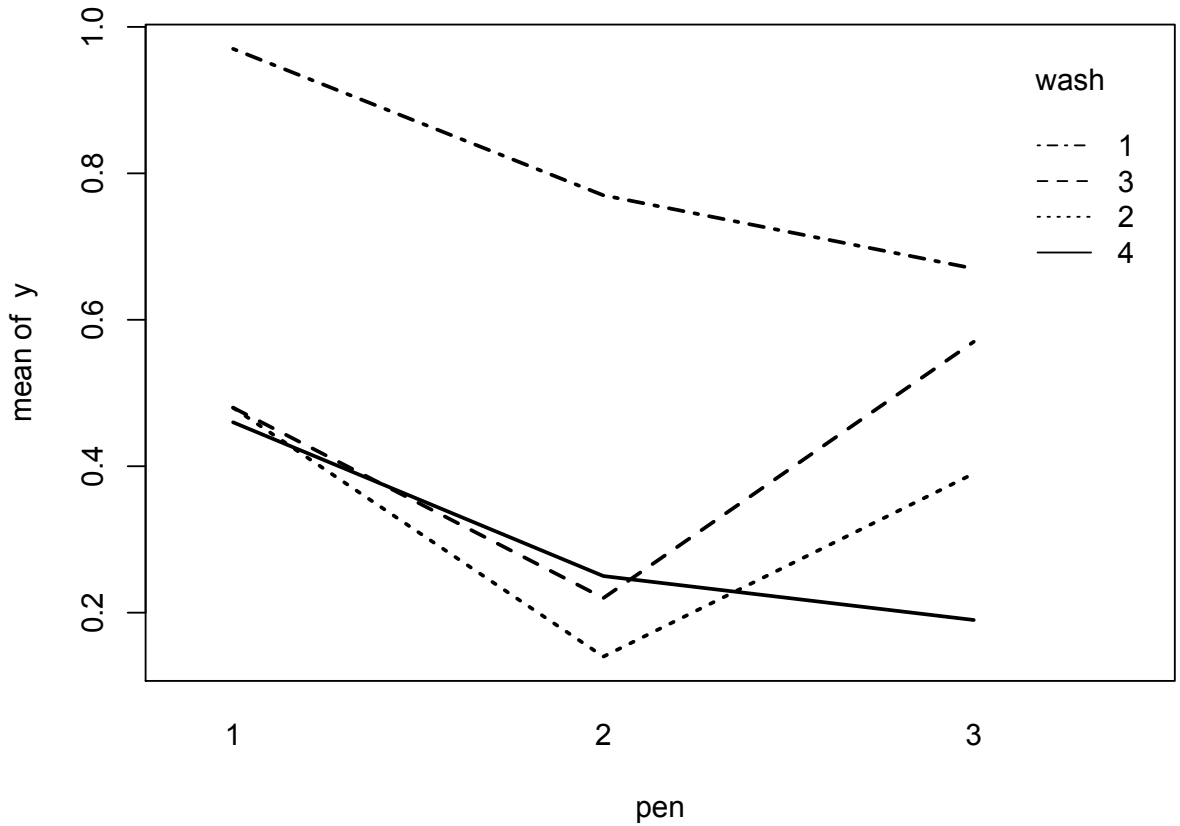
Two-factor ANOVA

```
> interaction.plot(wash, pen,y)
```



Two-factor ANOVA

```
> interaction.plot(pen, wash, y)
```



Two-factor ANOVA

In addition to H_{A+B} we consider the following sub-models:

$$H_A : \mu_{ij} = \alpha_i.$$

$$H_B : \mu_{ij} = \beta_j.$$

$$H_1 : \mu_{ij} = \mu_0.$$

We introduce the following notation:

$$\overline{Y}_{i\bullet} = \frac{1}{s}\sum_{j=1}^{s} Y_{ij}.$$

$$\overline{Y}_{\bullet j} = \frac{1}{r}\sum_{i=1}^{r} Y_{ij}.$$

$$\overline{Y}_{\bullet\bullet} = \frac{1}{rs}\sum_{i=1}^{r}\sum_{j=1}^{s} Y_{ij}.$$

Embedding into LM framework

Next we consider the subspaces corresponding to the linear hypotheses.

Obviously, $L_1 = \mathrm{span}(1_n)$. The spaces L_A and L_B can be deduced from the one-way ANOVA setup.

$$L_A = \mathrm{span}(e_1, \ldots, e_r), \quad \text{and} \quad L_B = \mathrm{span}(v_1, \ldots, v_s),$$

where

$$e_i = (\underbrace{0, \ldots, 0}_{s\times}, \ldots, \underbrace{0, \ldots, 0}_{s\times}, \underbrace{1, \ldots, 1}_{\substack{s\times \\ i\text{-th block}, \quad 1\le i\le r}}, \underbrace{0, \ldots, 0}_{s\times}, \ldots, \underbrace{0, \ldots, 0}_{s\times})',$$

and

$$v_j = (\underbrace{0, \ldots, 0, 1, 0, \ldots, 0}_{\substack{\text{length } s \\ 1 \text{ at position } j, \quad 1 \le j \le s}}, \ldots, \underbrace{0, \ldots, 0, 1, 0, \ldots, 0}_{\substack{\text{length } s \\ 1 \text{ at position } j, \quad 1 \le j \le s}})'.$$

Embedding into LM framework

We have

- $\dim(L_A) = r$, $\dim(L_B) = s$ and $\dim(L_1) = 1$;
- with $p_i = p_{L_i}$

$$(p_A(Y))_{ij} = \overline{Y}_{i\bullet}, \quad (p_B(Y))_{ij} = \overline{Y}_{\bullet j} \quad \text{and} \quad (p_1(Y))_{ij} = \overline{Y}_{\bullet\bullet};$$

- and

$$\hat{\alpha}_i = \overline{Y}_{i\bullet}, \quad \hat{\beta}_j = \overline{Y}_{\bullet j} \quad \text{and} \quad \hat{\mu}_0 = \overline{Y}_{\bullet\bullet}.$$

Embedding into LM framework

Now we investigate L_{A+B}. To this end we need the following lemma.

Lemma

Let $L_A^* = L_A \ominus L_1$ and $L_B^* = L_B \ominus L_1$. Denote with p_i^* the corresponding projections. Then:

1. $L_A^* = \{\mu \in \mathbb{R}^n : \mu_{ij} = \alpha_i \quad \text{and} \quad \sum_{i=1}^r \alpha_i = 0\}$ *and* $L_B^* = \{\mu \in \mathbb{R}^n : \mu_{ij} = \beta_j \quad \text{and} \quad \sum_{j=1}^s \beta_j = 0\}$.
2. $L_A^* \perp L_B \quad \text{and} \quad L_B^* \perp L_A$.
3. $L_{A+B} = L_A^* \oplus L_B^* \oplus L_1 \quad \text{and} \quad \dim(L_{A+B}) = r + s - 1$.
4. $p_{A+B} = p_A + p_B - p_1$.

Embedding into LM framework

Proof of 1. Note that if $\mu \in L_A^*$ then $\mu \in L_A$ and $\mu \perp 1_n$ hence

$$\mu' 1_n = s(\alpha_1 + \cdots + \alpha_r) = 0.$$

Proof of 2. Take $x \in L_A^*$ and $y \in L_B$. Then

$$x = (\alpha_1, \ldots, \alpha_1, \alpha_2, \ldots, \alpha_2, \ldots, \alpha_r, \ldots, \alpha_r)'$$

with $\sum \alpha_i = 0$ and

$$y = (\beta_1, \ldots, \beta_s, \beta_1, \ldots, \beta_s, \ldots, \beta_1, \ldots, \beta_s)'.$$

We get

$$x'y = (\alpha_1 + \cdots + \alpha_r)(\beta_1 + \cdots + \beta_s) = 0.$$

Embedding into LM framework

Proof of 3.

$$L_{A+B} = L_A + L_B = (L_A \ominus L_1) + (L_B \ominus L_1) + L_1 = L_A^* + L_B^* + L_1.$$

Now we use that the spaces L_A^*, L_B^* and L_1 are mutually orthogonal. Moreover, note that

$$\dim(L_{A+B}) = \underbrace{\dim(L_1)}_{1} + \underbrace{\dim(L_A^*)}_{r-1} + \underbrace{\dim(L_B^*)}_{s-1}.$$

Proof of 4. Noting that $p_A^* = p_A - p_1$ and $p_B^* = p_B - p_1$ the proof is immediate from 3.

Embedding into LM framework

With this lemma (using 4.) we can compute

$$(p_{A+B}(Y))_{ij} = \overline{Y}_{i\bullet} + \overline{Y}_{\bullet j} - \overline{Y}_{\bullet\bullet}.$$

Alternatively (using 3.) we get

$$(p_{A+B}(Y))_{ij} = \underbrace{\overline{Y}_{\bullet\bullet}}_{(p_1(Y))_{ij}} + \underbrace{(\overline{Y}_{i\bullet} - \overline{Y}_{\bullet\bullet})}_{(p_A^*(Y))_{ij}} + \underbrace{(\overline{Y}_{\bullet j} - \overline{Y}_{\bullet\bullet})}_{(p_B^*(Y))_{ij}}.$$

If we parametrize the model

$$\mu_{ij} = \mu_0 + \alpha_i^0 + \beta_j^0 \quad \text{with} \quad \sum_{i=1}^{r} \alpha_i^0 = \sum_{j=1}^{s} \beta_j^0 = 0,$$

then $\hat{\alpha}_i^0 = \overline{Y}_{i\bullet} - \overline{Y}_{\bullet\bullet}$ and $\hat{\beta}_j^0 = \overline{Y}_{\bullet j} - \overline{Y}_{\bullet\bullet}$.

Embedding into LM framework

Using our theory we can now estimate σ^2 under H_{A+B} as follows:

$$\tilde{\sigma}^2_{A+B} = \|Y - p_{A+B}(Y)\|^2/(n - \dim(L_{A+B})) = \frac{\text{SSR}}{(r-1)(s-1)} = \text{MSR}.$$

Here we used

$$n = rs \quad \text{and} \quad \dim(L_{A+B}) = r + s - 1.$$

Note that $E\tilde{\sigma}^2_{A+B} = \sigma^2$.

Testing

Now that we have embedded our model to the LM framework, we can easily derive the F-Statistics for different tests.

We consider first

$$H_A = H_{0B}(\text{no factor effect B}) \quad \text{under} \quad H_{A+B}.$$

Then

$$F(Y) = F^{0B}(Y) = \frac{\|p_{A+B}(Y) - p_A(Y)\|^2/(\dim(L_{A+B}) - \dim(L_A))}{\tilde{\sigma}^2_{A+B}}.$$

By 4. (lemma above), we get

$$\|p_{A+B}(Y) - p_A(Y)\|^2 = \|p_B(Y) - p_1(Y)\|^2 = \sum_{j=1}^{s} r(\overline{Y}_{\bullet j} - \overline{Y}_{\bullet\bullet})^2 = SSB.$$

Moreover, $\dim(L_{A+B}) - \dim(L_A) = s - 1$.

Testing

Wrapping up, we get

$$F^B(Y) = \frac{SSB/(s-1)}{SSR/(s-1)(r-1)} = \frac{MSB}{MSR} \overset{H_{0B}}{\sim} F_{(s-1),(s-1)(r-1)}.$$

We reject H_{0B} at significance level α if

$$F^{0B}(Y) > F_{(s-1),(s-1)(r-1);1-\alpha}.$$

Testing

$$H_B = H_{0A} \quad \text{under} \quad H_{A+B}.$$

is analogous.

Sums of square decomposition

By 4. (from the lemma above) we obtain

$$\begin{aligned} Y - p_{A+B}(Y) &= Y - p_A(Y) - p_B(Y) + p_1(Y) \\ &= (Y - p_1(Y)) - (p_A(Y) - p_1(Y)) - (p_B(Y) - p_1(Y)). \end{aligned}$$

Thus

$$(Y - p_1(Y)) = \underbrace{(Y - p_{A+B}(Y))}_{p^{\perp}_{A+B}} + \underbrace{(p_A(Y) - p_1(Y))}_{p^{*}_{A}} + \underbrace{(p_B(Y) - p_1(Y))}_{p^{*}_{B}}.$$

So on the right hand side we project on spaces $L^{\perp}_{A+B}$, L^{*}_{A} and L^{*}_{B} which are mutually orthogonal. By Pythagoras' theorem it follows that

$$\begin{aligned} &\|Y - p_1(Y)\|^2 \\ &= \|Y - p_{A+B}(Y)\|^2 + \|p_A(Y) - p_1(Y)\|^2 + \|p_B(Y) - p_1(Y)\|^2 \end{aligned}$$

In short: $SST = SSR + SSA + SSB$.

Sums of square decomposition

The goodness-of-fit can be measured by

$$R^2 = \frac{SSA + SSB}{SST}.$$

When R^2 (which is in $[0, 1]$) is large (i.e. close to 1) it means that the residuals are small compared to the variations with respect to factors and A and B.
Alternatively it is common to consider the corrected R^2

$$R^2_{\text{corr}} := 1 - \frac{n-1}{(r-1)(s-1)}(1 - R^2)$$

which is accounting for the number of parameters.

Data example

We get back to our introductory data example. Running the AOV we get the following:

```
> mod=aov(y~pen+wash)
> summary(mod)
            Df Sum Sq Mean Sq F value Pr(>F)
pen          2 0.1282 0.06411   4.432 0.0658 .
wash         3 0.4797 0.15990  11.055 0.0074 **
Residuals    6 0.0868 0.01446
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 '
```

The factor B (wash) is highly significant.

Data example

Alternatively we can use `lm()`. To make the results better interpretable we use the parametrization

$$\mu_{ij} = \mu_0 + \alpha_i^0 + \beta_j^0, \quad \sum \alpha_i^0 = \sum \beta_j^0 = 0.$$

In `R we can do this by defining contrasts:`

```
> contrasts(wash)="contr.sum"
> contrasts(pen)="contr.sum"
```

Data example

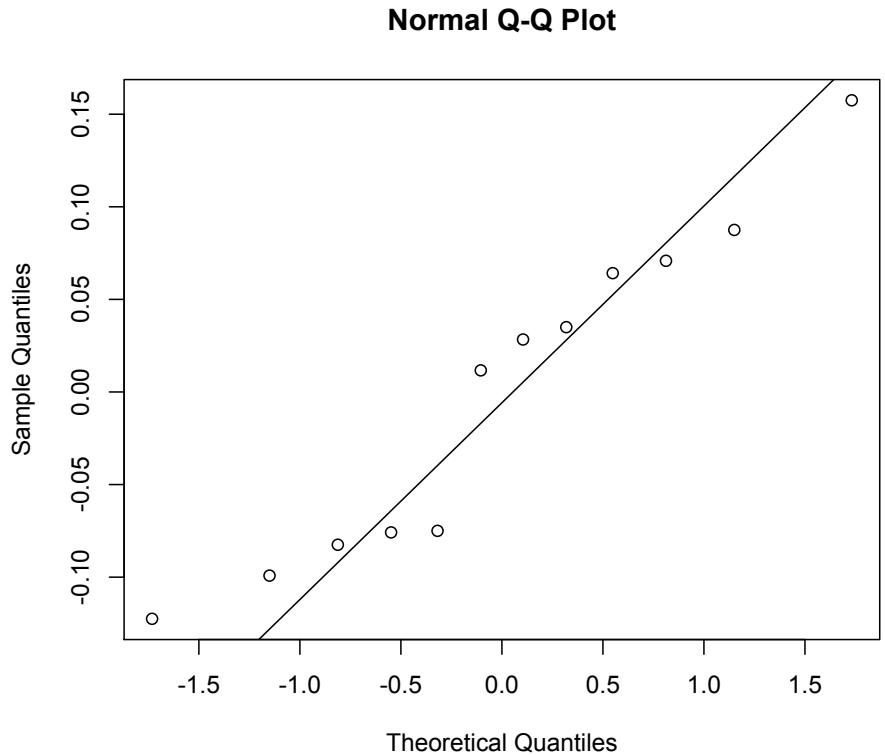
```
> mod=lm(y~pen+wash)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.46583    0.03472  13.418 1.06e-05 ***
pen1         0.13167    0.04910   2.682  0.03645 *
pen2        -0.12083    0.04910  -2.461  0.04905 *
wash1        0.33750    0.06013   5.613  0.00137 **
wash2       -0.12917    0.06013  -2.148  0.07531 .
wash3       -0.04250    0.06013  -0.707  0.50622
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1203 on 6 degrees of freedom
Multiple R-squared:  0.8751,Adjusted R-squared:  0.771
F-statistic: 8.406 on 5 and 6 DF,  p-value: 0.01104
```

Data example

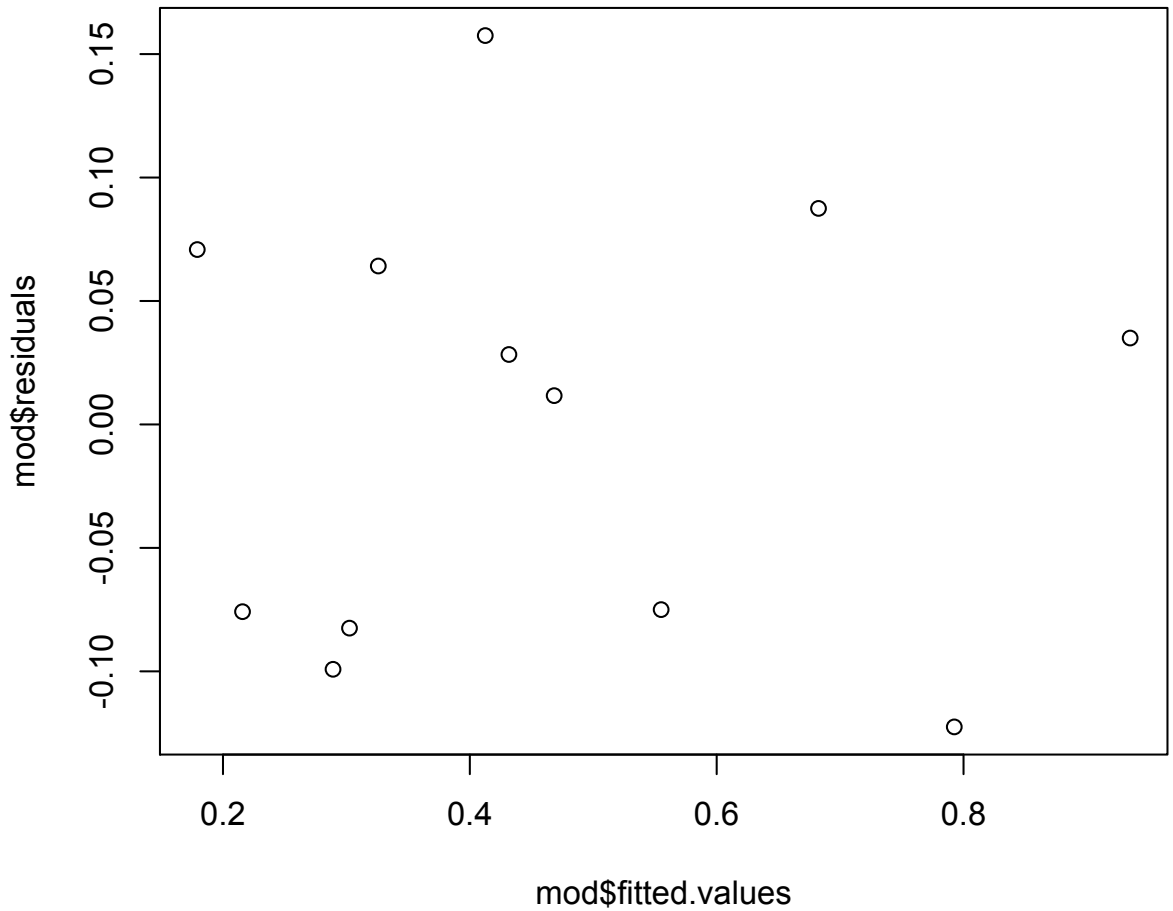
```
> qqplot(mod$residuals)
> qqline(mod$residuals)
```



There is no apparent deviation from normal, though with the little number of observation this is hard to judge.

Data example

```
> plot(mod$fitted.values,mod$residuals)
```



Randomized block models

The design with with 1 observation per factor combination appears in so-called randomized block models. Let us motivate in what context they appear.

Suppose we are interested in the effect of some factor A with r levels (treatment factor) on our responses. So in principle we are in the setting of a one-factor AOV. However, we conjecture that there are other influences on our responses beyond this single factor. In this case the effect of factor A may be confounded with potential other influences. More concretely, suppose that we want to test 3 different fertilisers (factor A). To this end we apply the fertilisers to 15 acres. If the acres were homogenous (in terms of soil, humidity, sun, etc.) we would attribute each fertiliser to 5 randomly chosen acers and do a one-factor AOV. However, if the acres are very heterogenous, then a good or bad crop may rather be attributed to the specific acres.

Randomized block models

As a consequence, it is always desirable to choose experimental units as homogenous as possible. This can be achieved by so-called blocking.

In our example, we would devide the 15 acres into 5 as homogeneous as possible groups. Then each group consists of 3 units and we randomly assign one fertiliser to each of these units. This is the principle of a randomized block design. The factor which comes from forming the homogenous groups is called a blocking factor.

The analysis can be carried out in exactly the way we have explained above.

The latin square

In common applications we can have more than one blocking factor. Say A and B are block factors and C is a treatment factor.

Example

- Agriculture:

$$\text{block factors} \begin{cases} \text{Factor A: type of vegetable} \\ \text{Factor B: acre} \end{cases}$$

$$\text{treatment factor: Factor C} \Longrightarrow \text{fertiliser}$$

- Industry:

$$\text{block factors} \begin{cases} \text{Factor A: raw material} \\ \text{Factor B: working unit} \end{cases}$$

$$\text{treatment factor: Factor C} \Longrightarrow \text{production method}$$

The latin square

For a complete design (with one observation per factor combination) we need $n = rst$ experimental units.

We consider now the special case where $r = s = t$ and aim to design an experiment with only r^2 instead of r^3 experimental units.
Latin square:

- we assume no interaction between factors;
- each level of factor C appears once in each column and row:

	B_1	B_2	B_3	B_4
A_1	C_1	C_2	C_4	C_3
A_2	C_2	C_3	C_1	C_4
A_3	C_3	C_4	C_2	C_1
A_4	C_4	C_1	C_3	C_2

The latin square

For the latin square typically the levels C_1, C_2, ... are coded as A, B, ...

$$\begin{array}{cccc} A & B & D & C \\ B & C & A & D \\ C & D & B & A \\ D & A & C & D \end{array}$$

There are many such arrangements possible (12 for $r = 3$, 576 for $r = 4$...) A randomly chosen arrangement decides which factor combinations are used.

The latin square

We let $k = k(i, j)$ be the level of the treatment factor when the block factors are set to levels i and j.

The following hypotheses are considered:

$$
\begin{aligned}
H_{A+B+C} &: \mu_{ij} = \alpha_i + \beta_j + \gamma_{k(i,j)} \\
H_A &: \mu_{ij} = \alpha_i \\
H_B &: \mu_{ij} = \beta_j \\
H_C &: \mu_{ij} = \gamma_{k(i,j)} \\
H_1 &: \mu_{ij} = \mu_0.
\end{aligned}
$$

The latin square

The same type of arguments that we have used in the complete design with 2 factors yield:

- $L_{A+B+C} = L_A^* \oplus L_B^* \oplus L_C^* \oplus L_1$.
- $\dim(L_1) = 3r - 2$.
- We set

$$
\begin{aligned}
SSA &= \|p_A(Y) - p_1(Y)\|^2, \\
SSB &= \|p_B(Y) - p_1(Y)\|^2, \\
SSC &= \|p_C(Y) - p_1(Y)\|^2, \\
SSR &= \|Y - p_{A+B+C}(Y)\|^2, \\
SST &= \|Y - p_1(Y)\|^2.
\end{aligned}
$$

Then $SST = SSA + SSB + SSC + SSR$.

The latin square

- We have that $SSR \sim \sigma^2 \chi^2_{(r-2)(r-1)}$.
- We define $H_{0A}\colon \alpha_i = 0$ for all i. This hypothesis corresponds to $\mu \in L_B^* \oplus L_C^* \oplus L_1$. Since $L_{A+B+C} = L_A^* \oplus L_B^* \oplus L_C^* \oplus L_1$ and

$$L_{A+B+C} \ominus (L_B^* \oplus L_C^* \oplus L_1) = L_A^* = L_A \ominus L_1$$

  we have

$$F^A = \frac{MSA}{MSR} \sim F_{r-1,(r-1)(r-2)}.$$

- Analogously, we define F^B and F^C and the corresponding tests.

The latin square

In the following example (Lawson p. 131) we study the concentration of some drug in the blood. The question was whether the formulation (A=solution, B=tablet, C=capsule) has some impact on it.

Three volunteers took each formulation in succession with sufficient washout period (time between treatments in order to avoid dependencies.)

Since there may be large variation from subject to subject, subject was used as row blocking factor. (Levels 1,2,3),

Since absorbtion will vary from time to time for a particular subject, time was used as second blocking factor.

The latin square

	Period		
Subject	1	2	3
1	A 1186	B 642	C 1183
2	B 1135	C 1305	A 984
3	C 873	A 1426	B 1540

The latin square

```
> library("daewr")
> bioeqv
  Subject Period Treat  AUC
1       1      1     A 1186
2       1      2     B  642
3       1      3     C 1183
4       2      1     B 1135
5       2      2     C 1305
6       2      3     A  984
7       3      1     C  873
8       3      2     A 1426
9       3      3     B 1540
> mod=aov(AUC~Subject+Period+Treat,data=bioeqv)
> summary(mod)
            Df Sum Sq Mean Sq F value Pr(>F)
Subject      2 114264   57132   0.258  0.795
Period       2  45196   22598   0.102  0.907
Treat        2  15000    7500   0.034  0.967
Residuals    2 442158  221079
```

The latin square

From this data no significant difference in the AUC value w.r.t. treatment (formulation) is visible.

For illustrative purposes, we show the means for each treatment and each block factor and a Tukey multiple comparison.

```
> model.tables(mod,type="means")
Grand mean
1141.556
Subject
     1      2      3
1003.7 1141.3 1279.7
Period
     1      2      3
1064.7 1124.3 1235.7
Treat
     A      B      C
1198.7 1105.7 1120.3
```

The latin square

```
> TukeyHSD(mod,"Treat")
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = AUC ~ Subject + Period + Treat, data = bioeqv)

$Treat
         diff       lwr      upr     p adj
B-A -93.00000 -2354.511 2168.511 0.9686678
C-A -78.33333 -2339.844 2183.178 0.9775653
C-B  14.66667 -2246.844 2276.178 0.9991960
```

The latin square

In the following example (Montgomery p. 194–198) we study the power of dynamite.

Five different piles of raw material (block factor A), five machinists (block factor B) and five production techniques (treatment factor C) were taken into account.

Pile of raw material	Machinist					Effects
	1	2	3	4	5	$\bar{y}_i^A - \bar{y}$
1	A=24	B=20	C=19	D=24	E=24	-3.2
2	B=17	C=24	D=30	E=27	A=36	+1.4
3	C=18	D=38	E=26	A=27	B=21	+0.6
4	D=26	E=31	A=26	B=23	C=22	+0.2
5	E=22	A=30	B=20	C=29	D=31	+1.0
$\bar{y}_j^B - \bar{y}$	-4.0	+3.2	-1.2	+0.6	+1.4	$\bar{y} = 25.4$
$\bar{y}_k^C - \bar{y}$	+3.2	-5.2	-3.0	+4.4	+0.6	

The latin square

This is how we get the design and data.frame.

```
library(agricolae)
Method=c("A", "B","C","D","E")
design=design.lsd(Method, randomization=FALSE)
datafr=design$book
colnames(datafr)<-c("y","Pile","Machinist","Method")
datafr$y=c(24,20,19,24,24,17,24,30,27,36,18,38,
26,27,21,26,31,26,23,22,22,30,20,29,31)
```

The latin square

The treatment factor Method appears significant.

```
> mod1=aov(y~Pile+Machinist+Method,data=datafr)
> summary(mod1)
            Df Sum Sq Mean Sq F value  Pr(>F)
Pile         4     68   17.00   1.594 0.23906
Machinist    4    150   37.50   3.516 0.04037 *
Method       4    330   82.50   7.734 0.00254 **
Residuals   12    128   10.67
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The latin square

```
contrasts(datafr$Method)=contr.sum
contrasts(datafr$Pile)=contr.sum
contrasts(datafr$Machinist)=contr.sum
lmod=lm(y~Pile+Machinist+Method,data=datafr)
summary(lmod)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  25.4000     0.6532  38.886 5.39e-14 ***
Pile1        -3.2000     1.3064  -2.449  0.03062 *
Pile2         1.4000     1.3064   1.072  0.30496
Pile3         0.6000     1.3064   0.459  0.65424
...
Method2      -5.2000     1.3064  -3.980  0.00182 **
Method3      -3.0000     1.3064  -2.296  0.04046 *
Method4       4.4000     1.3064   3.368  0.00559 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.266 on 12 degrees of freedom
Multiple R-squared:  0.8107,Adjusted R-squared:  0.6213
F-statistic: 4.281 on 12 and 12 DF,  p-value: 0.008855
```

The latin square

To get all effects in additive way, we need to set contrasts to `contr.sum` for all factors (which we already did above).

```
mod.addeffects=aov(y~Pile+Machinist+Method,data=datafr)
tab=model.tables(mod.addeffects)
tab
Tables of effects
Pile
   1    2    3    4    5
-3.2  1.4  0.6  0.2  1.0
Machinist
   1    2    3    4    5
-4.0  3.2 -1.2  0.6  1.4
Method
   A    B    C    D    E
 3.2 -5.2 -3.0  4.4  0.6
```

The latin square

With so-called profileplots we can visualize these effects.

```
> plot(tab[[1]]$Pile,ylab="Pile effects",type="o",lim=c(-5,5))
> abline(h=0)
> ...
```

